

- 1 (a) spacing = 380 or 3.8×10^2 pm B1 [1]
- (b) time = 24×3600
time = 0.086 (0.0864) Ms B1 [1]
- (c) time = distance / speed = $\frac{1.5 \times 10^{11}}{3 \times 10^8}$ C1
= 500 (s) = 8.3 min A1 [2]
- (d) momentum and weight B1 [1]
- (e) (i) arrow to the right of plane direction (about 4° to 24°) B1 [1]
- (ii) scale diagram drawn
or use of cosine formula $v^2 = 250^2 + 36^2 - 2 \times 250 \times 36 \times \cos 45^\circ$
or resolving $v = [(36 \cos 45^\circ)^2 + (250 - 36 \sin 45^\circ)^2]^{1/2}$ C1

resultant velocity = 226 (220 – 240 for scale diagram) ms^{-1}
allow one mark for values 210 to 219 or 241 to 250 ms^{-1}
or use of formula ($v^2 = 51068$) $v = 230$ (226) ms^{-1} A1 [2]
- 2 (a) (i) accelerations (A to B and B to C) are same magnitude B1